THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If x : y = 2 : 3 and y : z = 4 : 7, then x : y : z is: 1

(a) 2 : 4 : 7 **(b)** 8 : 12 : 21 **(c)** 2 : 7 : 7 **(d)** 2 : 3 : 7

2. The present ages of two brothers are in the ratio 3 : 2. After 4 years the ratio becomes 7 : 5. The present age of the elder brother is: 1

(a) 28 years **(b)** 40 years **(c)** 16 years **(d)** 24 years

3. ₹ 1300 is divided in parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. The largest part is: 1

(a) ₹ 260 **(b)** ₹ 600 **(c)** ₹ 300 **(d)** ₹ 400

4. 12 men can finish a job in 20 days. After 8 days, 4 more men join. The remaining work will be finished in: 1

(a) 9 days **(b)** 15 days **(c)** 11 days **(d)** 12 days

5. Two trains cover the same distance at speeds in the ratio 4 : 5. The ratio of the times they take is: 1

(a) 9 : 4 **(b)** 16 : 25 **(c)** 5 : 4 **(d)** 4 : 5

6. If 3 : 5 = 12 : 20, then (3+5) : 5 = x : 20, where x is: 1

(a) 37 **(b)** 20 **(c)** 12 **(d)** 32

7. The sides of two squares are in the ratio 2 : 3. The ratio of their areas is: 1

(a) 4 : 9 **(b)** 2 : 9 **(c)** 2 : 3 **(d)** 4 : 3

8. The number x for which 4 : 9 = 12 : x is: 1

(a) 21 **(b)** 27 **(c)** 25 **(d)** 18

9. **Assertion (A):** The ratios $3 : 4$ and $9 : 12$ are in proportion. 1

Reason (R): Two ratios $a : b$ and $c : d$ are in proportion if they are equal, that is, if $ad = bc$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The ratio $15 : 25$ in its simplest form is $3 : 5$. 1
Reason (R): The ratio of two quantities of the same kind has no unit.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The sides of two squares are in the ratio 3 : 7. Find the ratio of their perimeters and of their areas. 2

12. If x : y = 7 : 3, find the ratio (x + y) : (x - y). 2

13. Divide ₹ 1200 into parts proportional to $\frac{1}{2}$ and $\frac{1}{3}$. 2

14. The sides of two squares are in the ratio 4 : 5. Find the ratio of their perimeters and of their areas. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The ratio of the present ages of a father and son is 3 : 1. After 4 years the ratio will be 7 : 3. Find their present ages. 3

16. 15 men can finish a job in 24 days. After 8 days, 5 more men join them. In how many more days is the job finished? 3

17. Divide ₹ 2600 into three parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. 3

OR

The ratio of the present ages of a father and son is 5 : 2. After 6 years the ratio will be 2 : 1. Find their present ages. 3

18. 10 men can finish a job in 21 days. After 6 days, 5 more men join them. In how many more days is the job finished? 3

OR

Divide ₹ 3800 into three parts proportional to $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{5}$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita in Amritsar mixes curd and water in the ratio $3 : 2$ to make 10 litres of lassi. 5
(i) What fraction of the lassi is curd?
(ii) How many litres of curd are used?
(iii) How many litres of water are used?
(iv) She wants to make 25 litres of lassi in the same ratio. Find the quantities of curd and water and show that the ratio is unchanged.

20. A, B and C invest ₹ 3000, ₹ 2000 and ₹ 4000 for 4, 6 and 2 months respectively. A profit of ₹ 1200 is shared in the ratio of (investment $\times$ time). Find the ratio and each share. 5

OR

A vessel holds 90 L of a mixture of milk and water in the ratio 5 : 4. 18 L of the mixture is removed and replaced by 18 L of water. Find the new ratio of milk to water. 5

END OF QUESTION PAPER Total: 40 Marks